

Operating System

4th Semester, Academic Year 2025

Date:05-03-2025

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Unit#____2_____

Program Number: 2

Q2. Demo shared memory.

- One program will create a shared memory segment and write the student details (name, SRN, sem, CGPA) to it
- Other program will read from a shared memory segment and display student information

CODE:

SENDER.C

```
#include <sys/ipc.h>
#include <sys/shm.h>
#include <stdio.h>
#include <unistd.h>
#include <string.h>
```

```
struct Student {
    char name[50];
```

```

    char SRN[20];
    int sem;
    float CGPA;
};

int main() {
    void *shared_memory;
    int shmid;

    shmid = shmget((key_t)1122, sizeof(struct Student), 0666 |
IPC_CREAT);
    printf("Key of shared memory is %d\n", shmid);

    shared_memory = shmat(shmid, NULL, 0); //attach
    printf("Process attached at %p\n", shared_memory);

    struct Student *student = (struct Student *)shared_memory;

    // input
    printf("Enter student name: ");
    fgets(student->name, sizeof(student->name), stdin);
    student->name[strcspn(student->name, "\n")] = 0; // newline

    printf("Enter SRN: ");
    scanf("%s", &student->SRN);

    printf("Enter semester: ");
    scanf("%d", &student->sem);

    printf("Enter CGPA: ");
    scanf("%f", &student->CGPA);

    printf("\nData written to shared memory.\n");

    return 0;
}

```

RECIEVER.C

```
#include <sys/ipc.h>
#include <sys/shm.h>
#include <stdio.h>
#include <unistd.h>
#include <string.h>
```

```
struct Student {
    char name[50];
    char SRN[20];
    int sem;
    float CGPA;
};
```

```
int main() {
    void *shared_memory;
    int shmid;
```

```
    shmid = shmget((key_t)1122, sizeof(struct Student), 0666);
    printf("Key of shared memory is %d\n", shmid);
```

```
    shared_memory = shmat(shmid, NULL, 0);
    printf("Process attached at %p\n", shared_memory);
```

```
    struct Student *student = (struct Student *)shared_memory;
```

```
    printf("\nStudent Information:\n");
    printf("Name: %s\n", student->name);
    printf("SRN: %s\n", student->SRN);
    printf("Semester: %d\n", student->sem);
    printf("CGPA: %.2f\n", student->CGPA);
```

```
    return 0;
}
```

OUTPUT:

```
rithvikmatta@penguin:~/sem4/OS/banana prob 2$ gcc sender.c
rithvikmatta@penguin:~/sem4/OS/banana prob 2$ ./a.out
Key of shared memory is 43
Process attached at 0x7f8804ac9000
Enter student name: Rithvik Rajesh Matta
Enter SRN: PES2UG23CS485
Enter semester: 4
Enter CGPA: 8.37

Data written to shared memory.
rithvikmatta@penguin:~/sem4/OS/banana prob 2$ gcc reciever.c
rithvikmatta@penguin:~/sem4/OS/banana prob 2$ ./a.out
Key of shared memory is 43
Process attached at 0x7fe640c29000

Student Information:
Name: Rithvik Rajesh Matta
SRN: PES2UG23CS485
Semester: 4
CGPA: 8.37
```

If found plagiarized, I will abide with the disciplinary action of the University.

Signature: Rithvik Rajesh Matta

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